

Telegram @satuzbhunters

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A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the base of each of the two pyramids is 100 cm^2 . Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2 ?

- To study fluctuations in the leaf water potential, samples of wood were taken from 26 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 2.00 centimeters. If this cube has a mass of 5.12 grams, what is the density of this cube in grams per cubic centimeter?

The figure shown consists of a rectangle and a semicircle, where the length of $\overline{WX}$ is 20 units and the length of $\overline{WZ}$ is 5 units. The diameter of the semicircle is $\overline{WX}$. Which of the following expressions represents the area, in square units, of the figure?

- A) $20 + 5 + 10\pi$ C) $20 \times 5 + 50\pi$
B) $20 \times 5 + 10\pi r$ D) $20 \times 5 + 100\pi$

4.

Square P has a side length of x inches. Square Q has a perimeter that is 32 inches greater than the perimeter of square P . The function f gives the area of square Q , in square inches. Which of the following defines f ?

- A) $f(x) = (x + 8)^2$
- B) $f(x) = (x + 32)^2$
- C) $f(x) = (32x + 8)^2$
- D) $f(x) = (32x + 32)^2$

5.

	Volume (cubic units)
Right circular cylinder A	112π
Right circular cylinder B	$3,024\pi$

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 4 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

6.

To estimate the proportion of a population that has a certain characteristic, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the characteristic is 0.2, with an associated margin of error of 0.08. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the characteristic?

- A) It is plausible that the proportion is between 0.12 and 0.28.
- B) It is plausible that the proportion is less than 0.12.
- C) The proportion is exactly 0.2.
- D) It is plausible that the proportion is greater than 0.28.

7.

A researcher is designing a study to investigate the average number of hours a student at a high school spends reading per day. The researcher will report the average number of hours a student at the high school spends reading with an associated margin of error. The researcher is considering using a random sample of either 105 or 210 students from the high school. Which of the following could be the most likely effect of using the larger random sample compared to the smaller random sample?

- A) The reported margin of error would be lower.
- B) The reported margin of error would be higher.
- C) The reported average number of hours would be lower.
- D) The reported average number of hours would be higher.

8.

A proposal for a construction project was included on a city election ballot. A TV news report stated that 8 times as many people voted in favor of the proposal as people who voted against it. An internet article reported that 24,500 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

9.

In a random sample of 276 residents in a certain building, 9 residents reported that they take the train to work. If there are a total of 1,380 residents in this building, which of the following is the best estimate of the number of building residents who take the train to work?

- | | |
|-------|----------|
| A) 9 | C) 276 |
| B) 45 | D) 1,095 |

10.

The values in data sets X and Y are shown in the table.

Data set X	13	13	14	14	15	16	16	17	17
Data set Y	2	2	3	3	4	5	5	6	6

The standard deviation of data set X is g , and the standard deviation of data set Y is s . Which of the following statements about the standard deviation of the data sets is true?

- A) $g < s$
- B) $g > s$
- C) $g = s$
- D) The relationship between g and s cannot be determined.

11.

Value	Data set A frequency	Data set B frequency
40	2	8
43	4	7
46	5	5
49	7	4
52	8	2

Data set A and data set B each consist of 26 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

- A) The mean of data set A is greater than the mean of data set B .
- B) The mean of data set A is less than the mean of data set B .
- C) The mean of data set A is equal to the mean of data set B .
- D) There is not enough information to compare the means of the data sets.

12.

Dataset A : 2, 2, 2, 5, 5, 8, 11, 11, 14, 14, 14

Dataset B : 2, 5, 5, 8, 8, 8, 8, 11, 11, 14

Data set A and data set B each have 11 values. Which of the following statements best compares the median of data set A and the median of data set B ?

- A) The median of data set A is greater than the median of data set B .
- B) The median of data set A is less than the median of data set B .
- C) The medians of data sets A and B are equal
- D) There is not enough information to compare the medians.

13.

Data set F consists of 55 integers, where the value of each integer is between 150 and 260. Data set G consists of the same 55 integers in F as well as the integers 10. Which of the following must be less for data set G than for data set F ?

- I. The mean
 - II. The median
- A) I only
B) II only
C) I and II
D) Neither I nor II

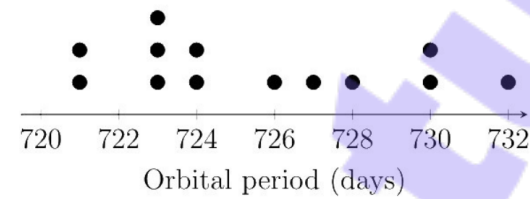
14.

Number of queen ants	Number of colonies
1	11
2	9
3	7
4	6
5	6

Scientists collected acorns that each housed a colony of a particular species and then analyzed each colony's structure. For these colonies, the table summarizes the number of queen ants per colony. For these colonies, what is the median number of queen ants per colony?

- A 6 C 3
B 4 D 2

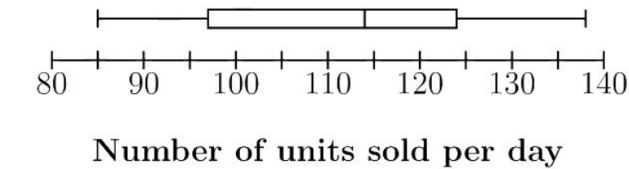
15.



A data set of the orbital periods, rounded to the nearest whole number of Earth days for 13 of Jupiter's moons is represented in the dot plot. An additional moon with an orbital period of 628 days is added to the original data set to create a new data set of 14 orbital periods. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

- A) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is equal to the median of the original data set.
- B) The mean of the new data set is equal to the mean of original data set, and the median of the new data set is less than the median of the original data set.
- C) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- D) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

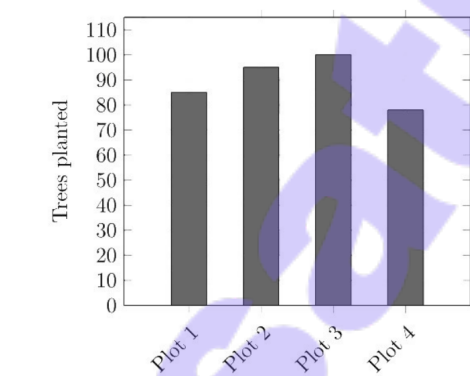
16.



A store owner recorded the number of units of a certain product that were sold per day for 28 days. What is the minimum number of units of this product sold per day for these 28 days?

17.

The bar graph shows the number of trees planted for 4 separate plots of land on a tree farm.



What is the maximum number of trees planted on one of these 4 plots of land?

- A) 100
- B) 95
- C) 85
- D) 78

18.

A researcher surveyed undergraduate students, graduate students, and postgraduate students. The number of undergraduate students surveyed was 7450% of the number of postdoctoral students surveyed and the number of graduate students surveyed was 24% of the undergraduate students surveyed. If there were 3576 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

19.

The positive number a is 2,352% of the sum of the positive numbers b and c , and b is 84% of c . What percent of b is a ?

- A) 5,152% C) 51.52%
B) 2,436% D) 24.36%

20.

The mass of object A is 498% of the mass of object B, and the mass of object A is 0.830% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

21.

For the positive quantities h , j , and k . 94% of h is equivalent to 47% of j , and j is equivalent to 30% of k . What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if you answer is 39%, enter 39)

22.

A square map has a side length of 35 inches, and 1 inch on the map represents an actual distance of 17 miles. A smaller version of the same map is printed as a square with the side length 55% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

- | | |
|----------|----------|
| A) 7.65 | C) 19.25 |
| B) 10.97 | D) 37.78 |